



CSCI 39579

Introduction to Excel

Excel is used to:



1. Store and organize data,
2. Analyze data, *and*
3. Represent data graphically (e.g., in bar graphs, histograms, and scatterplots)

Excel Basics



- Microsoft Excel consists of Workbooks
- Each Workbook is made up of an infinite number of worksheets
- It is possible to name each worksheet to aid in organizing your data

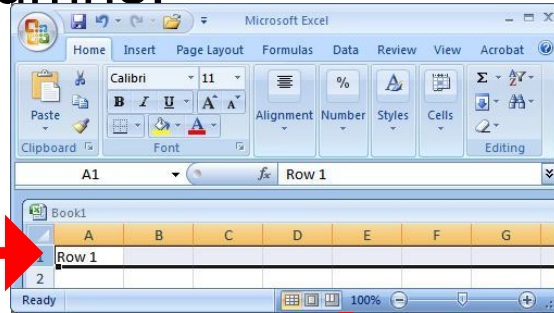
Excel Basics

Excel spreadsheets organize information (text and numbers) by rows and columns:



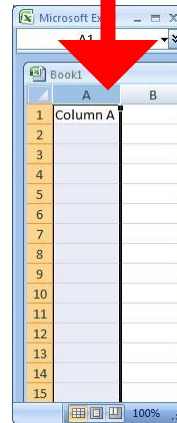
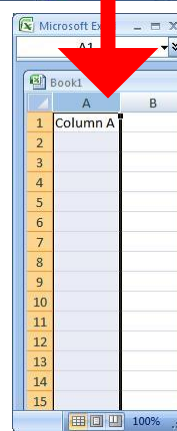
This is a **row**.

Rows are represented by **numbers** along the side of the sheet.



This is a **column**.

Columns are represented by **letters** across the top of the sheet.

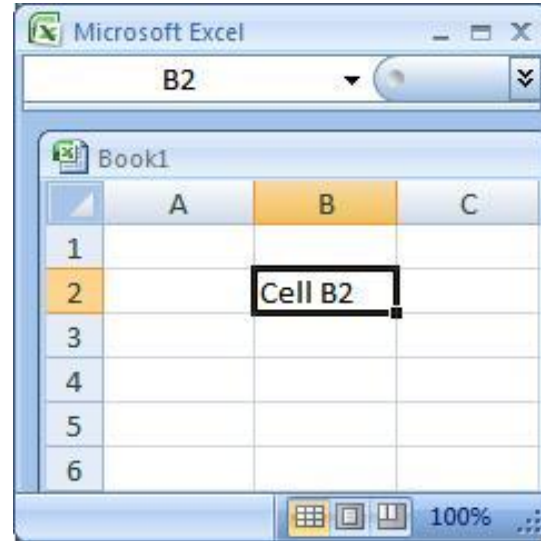


Excel Basics

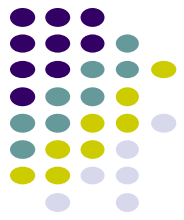


A **cell** is the intersection between a column and a row.

Each cell is named for the column letter and row number that intersect to make it.



Excel Basics

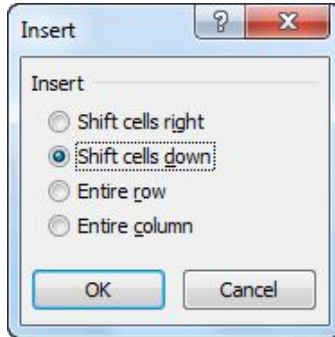


- Excel allows for some basic actions which we have used before in other programs
- Copy/Paste (Rows, Columns, Cells, etc.)
- Insert (Rows and Columns)
- Sort (Alphabetically, Numerically, Chronologically, etc.)

Inserting Rows and Columns



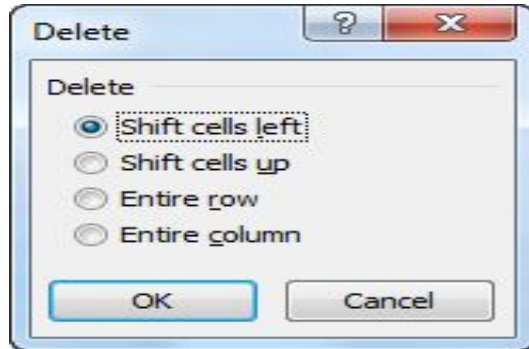
- The **Insert** command offers several techniques to insert rows, columns, and cells



Deleting Rows and Columns



- The **Delete** command offers several techniques to remove rows, columns, and cells



Formatting



- **Merging** allows us to combine two or more adjacent cells physically (disregarding contents)
 - To merge cells simply highlight the cells to be merged and select the merge option from the home menu
- Cells can also be formatted with options you are probably familiar with (bold, font size, borders, etc.)
 - These options can be found scattered on the home menu or under the format cells menu via the right-click

Formatting



- Excel also allows us to format cells by their data types. This is useful for a variety of reasons (sorting, manipulating, rounding, etc.)
- To format the cell's data type, in the numbers section of the home menu, select the number drop-down menu (defaults to General) and select the new type desired

- **Column width** is the horizontal measurement of a column

E8		Width: 11.86 (88 pixels)		fx		0.25	
	A	B	C	D	E	F	G
1	OK Office Systems Pricing Information						
2	1-Sep-12						
3							
4	Product	Cost	Markup R	Retail Price	Percent O	Sale Price	Profit Margin
5	Computer	475.5	0.5	713.25	0.15	606.2625	0.215686
6	Color Lase	457.7	0.755	803.2635	0.2	642.6108	0.287749
7	Filing Cab	68.75	0.905	130.9688	0.05	124.4203	0.447437
8	Desk Chai	75	1	150	0.25	112.5	0.333333
9	Solid Oak	700	1.857	1999.9	0.3	1399.93	0.499975
10	28" Monit	195	0.835	357.825	0.1	322.0425	0.39449
11							
12							

Adjusting Row Height



- **Row height** is the vertical measurement of a row
 - The row height is automatically adjusted with a font size increase
 - Using ALT+Enter to create multiple lines may require a row height adjustment
 - Select Row Height from the Format menu

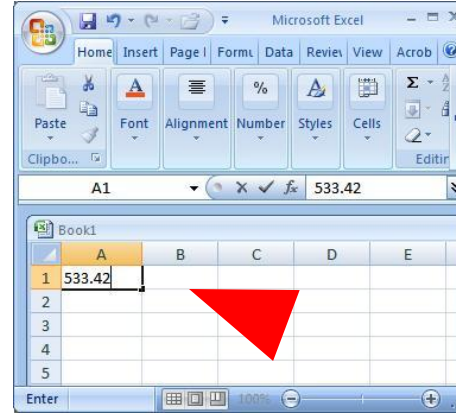
Data Entry



There are two ways to enter information into a cell:

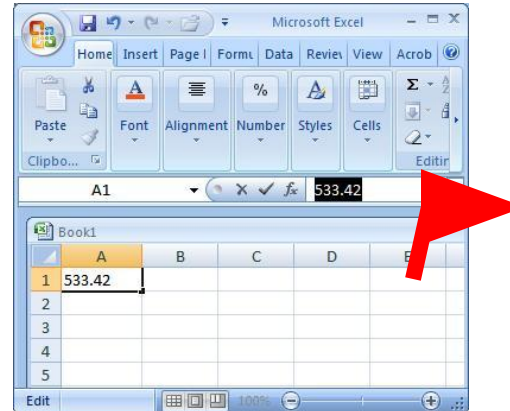
1. Type directly into the cell.

Click on a cell, and type in the data (numbers or text) and press Enter.



2. Type into the formula bar.

Click on a cell, and then click in the formula bar (the space next to the ). No  type the data into the bar and press Enter.





Entering and Editing Cell Data

- Excel supports text, values, dates, and formula results

	A	B	C	D	E	F
1						
2						
3	Text	Date	Value	Value	Formula Results	
4	↓	↓	↓	↓	↓	
5	Computer	9/1/2012	400	0.5	600	
6	Computer!					
7						
8						

Numeric Formats



Format Style	Display
General	A number as it was originally entered.
Number	A number with or without the 1,000 separator
Currency	A number with the 1,000 separator and with an optional dollar sign to the immediate left.
Accounting	A number with the 1,000 separator and with an optional dollar sign at the left cell border.
Date	The date in different ways, such as March 14, 2012 or 3/14/12.
Time	The time in different ways, such as 10:50 PM or 22:50 (24-hour time).

Numeric Formats Continued



Format Style	Display
Percentage	A value as it would be multiplied by 100 with the percent sign.
Fraction	A number as a fraction; appropriate when there is no exact decimal equivalent.
Scientific	A number as a decimal fraction followed by a whole number exponent of 10.
Text	The data left-aligned; is useful for numerical values that have leading zeros and should be treated as text.
Special	A number with editing characters, such as hyphens.
Custom	Predefined customized number formats or special symbols to create your own format.

Numeric Formats Continued



	A	B	C
1	General	1234.56	
2	Number	1234.56	
3	Currency	\$1,234.56	
4	Accounting	\$ 1,234.56	
5	Comma	1,234.56	
6	Percent	12.34%	
7	Short Date	3/1/2012	
8	Long Date	Thursday, March 01, 2012	

Data Removal



- Data can be removed from a cell, column, or row easily
- Here are a few methods:
 - Click the column or row heading that you want deleted, then click the Delete in the Cells group on the Home tab
 - Another method for entire row/column deletion is to click a cell in the row or column and follow the above method respectively
 - Delete methods are also found via the right-click menu
 - To remove data from a cell or group of cells, simply highlight those to be deleted and press delete

- [illegible]

[illegible]

Auto Fill



- Auto Fill enables us to copy the contents of a cell or a range of cells by dragging the *fill handle* over adjacent cells or a range
- To use Auto Fill:
 - 1. Click the cell with the content you want to copy to make it the active cell
 - 2. Position the pointer over the bottom-right corner of the cell until it changes to the fill pointer (a thin black plus sign)
 - 3. Drag the fill handle to repeat the content in other cells



Formulas and Functions

- Formulas are equations that perform calculations in your spreadsheet. Formulas always begin with an equals sign (=). When you enter an equals sign into a cell, you are basically telling Excel to “calculate this.”
- Functions are Excel-defined formulas. They take data you select and enter, perform calculations on them, and return value(s).



Formulas

Mathematical Symbols

Operation	Common Symbol	Symbol in Excel
Addition	+	+
Subtraction	-	-
Multiplication	X	*
Division	÷	/
Exponentiation	^	^

Order of Precedence



- **Order of precedence** (operations) controls the sequence in which math operators are computed
 - Parentheses
 - Exponentiation
 - Multiplication and Division
 - Addition and Subtraction

Cell References in Formulas



- It is best to use cell addresses in formulas versus actual data
 - If cell A1 contains value 5 and you need to add B1 to this value, use `=A1+B1` versus `=5+B1`
- If the data changes, Excel will recalculate the result

Functions



- All functions have a common format – the equals sign followed by the function name followed by the input in parentheses.
- The input for a function can be either:
 - A set of numbers (e.g., “=AVERAGE(2, 3, 4, 5)”)
 - This tells Excel to calculate the average of these numbers.
 - A reference to cell(s) (e.g., “=AVERAGE(B1:B18)” or “=AVERAGE (B1, B2, B3, B4, B5, B6, B7, B8)”)
 - This tells Excel to calculate the average of the data that appear in all the cells from B1 to B8.
 - You can either type these cell references in by hand or by clicking and dragging with your mouse to select the cells.

Functions for Descriptive Statistics



Below are several functions you may need to learn for this class.

=AVERAGE(first cell:last cell): calculates the mean

=MEDIAN(first cell:last cell): calculates the median

=MODE(first cell:last cell): calculates the mode

=VARP(first cell:last cell): calculates the variance

=STDEVP(first cell:last cell): calculates the standard deviation

- You may directly write the functions for these statistics into cells or the formula bar, OR
- You may use the function wizard ( in the toolbar)